Theorem 9: upper bound for the local edit distance

Proof overview of the Lean 4/Mathlib formalisation

Folder `GsimOT/Theorem9_LocalEditUpperBound`

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport.*
File `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository; statement, equation, page and line numbers refer to this PDF. `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

1 Statement

Paper. Theorem 9 (Section 3.2, p. 9, l. 297–299; proof in Appendix D.2, pp. 28–30, l. 864–947): *For all graphs G, H of the same order, $\delta_1(G,H) \le 15\sqrt{\delta_1^{\mathrm{OT}}(G,H)\cdot\ln\bigl(4/\delta_1^{\mathrm{OT}}(G,H)\bigr)}$, where the right-hand side is defined as 0 if $\delta_1^{\mathrm{OT}}(G,H)=0$.* This is the δ_1 bound of Theorem 2 (p. 3, l. 118–121). The remark after Theorem 9 (p. 9, l. 318–320) adds that it yields $\delta_1 \le C_\alpha(\delta_1^{\mathrm{OT}})^\alpha$ for every $\alpha < 1/2$.

Lean. For finite types V, W and simple graphs G on V, H on W, assuming `hCVZ : CVZRounding` (Lemma 30, the cited rounding theorem of [CVZ11]):

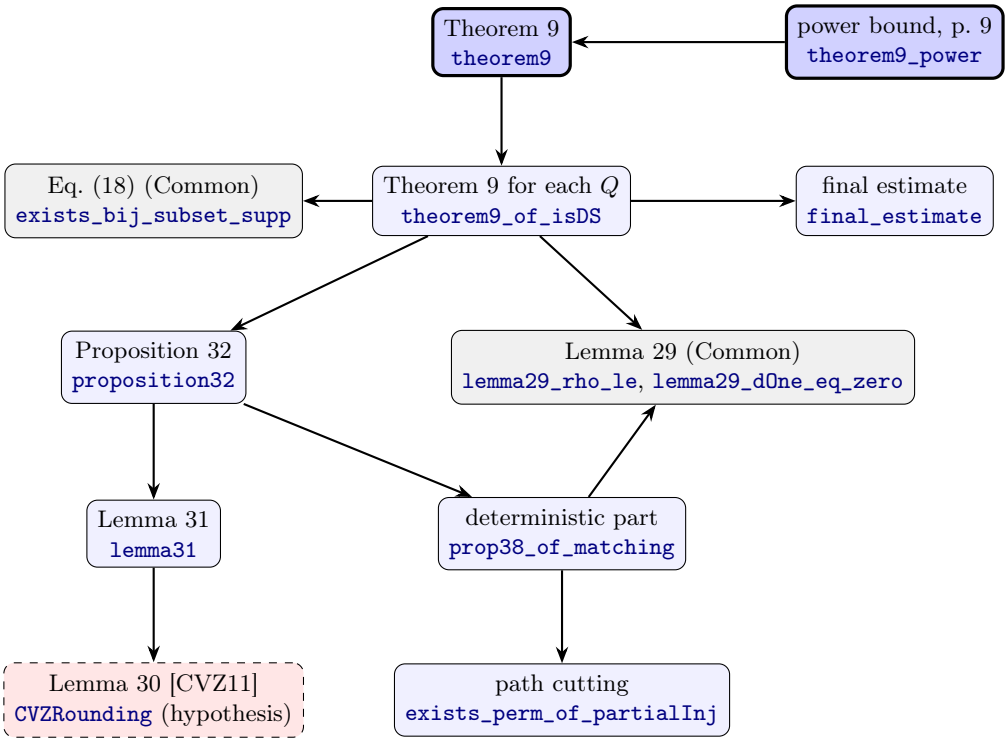
- `GsimOT.theorem9`: if $|V| = |W|$, then $\delta_1(G,H) \le 15\sqrt{\delta_1^{\mathrm{OT}}(G,H)\ln(4/\delta_1^{\mathrm{OT}}(G,H))}$.
- `GsimOT.theorem9_of_isDS`: the same with δ_1^{OT} replaced by $C_1(Q)/n$, for every $Q \in \mathrm{DS}(V,W)$.
- `GsimOT.theorem9_power`: for every $\alpha < 1/2$ there is C such that $\delta_1(G,H) \le C\,(\delta_1^{\mathrm{OT}}(G,H))^\alpha$ for all G, H of the same order.

In Lean $\ln 0 = 0$ and $4/0 = 0$, so the paper's convention for $\delta_1^{\mathrm{OT}} = 0$ is built in. Apart from the hypothesis `hCVZ`, the statement is exactly the paper's.

2 Where the proof lives

File	Content
`Lemma30_DependentRounding.lean`	Lemma 30 ([CVZ11, Thm. 1.1]) as the proposition `CVZRounding` (not proved; a hypothesis), matchings `IsMatching`.
`Lemma31_GoodMatching.lean`	$\Phi_v(M)$ (`errAt`); Lemma 31 (`lemma31`).
`PathCutting.lean`	The combinatorics of the proof of Proposition 32: cutting paths of a partial injection into blocks and closing them (`PathCutting.exists_perm_of_partialInj`).
`Proposition32_Rounding.lean`	Proposition 32: deterministic part `prop38_of_matching` and `proposition32`.
`Theorem9.lean`	Proof of Theorem 9 (`theorem9_of_isDS`, `final_estimate`, `theorem9`) and the power bound (`theorem9_power`).
`Common/Lemma29_Integrality.lean`	ρ_H and Lemma 29 (shared with Proposition 8).
`Common/Birkhoff.lean`	Eq. (18): a bijection σ inside $\mathrm{supp}(Q)$ (`exists_bij_subset_supp`).

3 Proof layout



Arrows mean "uses". Blue: this folder; grey: `GsimOT/Common`; red, dashed: cited result, an explicit hypothesis. `theorem9` also uses Lemma 29 directly for the isomorphic case.

4 The proof step by step

Setting, Eq. (18) (p. 28, l. 864–871). The paper fixes an optimal Q and $L = d_1^{\mathrm{OT}}$. The Lean works with an arbitrary $Q \in \mathrm{DS}(V, W)$ and $L := C_1(Q)$, so that all supported local costs are $\le L$ (`localCost_le_costOne`); the reference bijection $\sigma \subseteq \mathrm{supp}(Q)$ of Eq. (18) is `exists_bij_subset_supp`.

Lemma 29 (p. 28, l. 872–886). Shared with Proposition 8 and formalised in `GsimOT/Common`: $\rho_H(\sigma(v), w) \le 2L$ on $\mathrm{supp}(Q)$ (`lemma29_rho_le`), and $L < 1$ implies that σ is an isomorphism (`lemma29_iso`), hence $d_1 = 0$ (`lemma29_dOne_eq_zero`).

Φ_v and Lemma 30 (pp. 28–29, l. 887–898). $\Phi_v(M)$ is `errAt`. Lemma 30 is `CVZRounding`: for doubly stochastic Q and $\gamma \in (0, 1/2]$ a finitely supported distribution on matchings with marginals $(1-\gamma)Q$ and the tail bound (19) for every linear statistic with coefficients in $[0, 1]$.

Lemma 31 (p. 29, l. 899–907). `lemma31` follows the proof: $\mathbb{E}[\Phi_v(M)] \le L$; Eq. (19) with $\mu := L$, $\tau := 2 + 10\ln(4n)/(\gamma L)$ bounds each bad event by $1/(4n)$; the union bound gives $\le 1/4$; Markov's inequality gives $\Pr(n - |M| > 2\gamma n) \le 1/2$; so some matching in the support satisfies both bounds.

Proposition 32 (pp. 29–30, l. 908–935). `prop38_of_matching` takes any matching $M \subseteq \mathrm{supp}(Q)$ with $\Phi_v(M) \le X$ and shows $d_1 \le X + (n - |M|) + n/\ell + 2\ell L$. The partial injection $\mu(v) = \sigma^{-1}(w)$ for $(v, w) \in M$ is cut into blocks of ℓ vertices and closed into cycles (l. 912–921) by `PathCutting.exists_perm_of_partialInj`: a permutation τ that agrees with μ outside a set B of $\le m + n/\ell$ closing edges, each spanning $< \ell$ steps of μ; then $\pi := \sigma \circ \tau$. The claim $\rho_H(\sigma(v), \pi(v)) \le 2\ell L$ (l. 922–926) uses Lemma 29 along the block and the triangle inequality for ρ_H (`rho_triangle`). The final estimate of $L_{P_\pi}(v, \pi(v))$ (l. 927–935) inserts $A_H(\sigma(v), \pi(v'))$ and bounds the two sums by $\Phi_v(M) + m + n/\ell$ and $\rho_H(\sigma(v), \pi(v))$. `proposition32` combines this with Lemma 31.

Proof of Theorem 9 (p. 30, l. 936–947). In `theorem9_of_isDS`: if $L < 1$, $G \cong H$ by Lemma 29 and $\delta_1 = 0$. Otherwise $\ell := \lceil\sqrt{n/(2L)}\rceil$, $\gamma^* := \sqrt{5\ln(4n)/n}$ and $n/\ell + 2\ell L \le 2\sqrt{2nL} + 2L$; if $\gamma^* \le 1/2$, Proposition 32 gives Eq. (20) $d_1 \le 4L + 2\sqrt{2nL} + 4\sqrt{5n\ln(4n)}$, and if $\gamma^* > 1/2$, Eq. (20) follows from $d_1 \le n$ (`dOne_le_card`). Normalising uses that $x \mapsto \ln(4x)/x$ is decreasing for $x \ge 1$ (`log_four_mul_div_antitone`) with $n \ge 1/f \ge 1$. The final estimate $\frac{b}{\sqrt{fh}} \le 4\sqrt{f/h} + \frac{2\sqrt{2}}{\sqrt{h}} + 4\sqrt{5} \le \frac{4+2\sqrt{2}}{\sqrt{\ln 4}} + 4\sqrt{5} < 15$ is `final_estimate`. Finally `theorem9` passes from $C_1(Q)/n$ to the infimum δ_1^{OT} by continuity of $f \mapsto 15\sqrt{f\ln(4/f)}$ at $f > 0$ (`exists_lt_of_dOTOne_lt`); the case $d_1^{\mathrm{OT}} < 1$ is again the isomorphic case.

Power bound (p. 9, l. 320). `theorem9_power`: with $\varepsilon := 1 - 2\alpha > 0$, $\ln x \le x^\varepsilon/\varepsilon$ gives $\sqrt{f\ln(4/f)} \le \sqrt{4^\varepsilon/\varepsilon}\, f^\alpha$ (`sqrt_mul_log_le_rpow`), so $C_\alpha := 15\sqrt{4^\varepsilon/\varepsilon}$ works.

5 Assumptions and trusted base

- **Lemma 30 is a hypothesis**: `CVZRounding` is an explicit argument `hCVZ` of `theorem9`, `theorem9_of_isDS`, `theorem9_power`, `proposition32`, `lemma31`; no `axiom` is introduced. It is the paper's statement specialised to doubly stochastic Q (the only case used; a point of the perfect matching polytope), with "random matching" modelled as a finitely supported probability distribution on subsets of $V \times W$.
- Otherwise only `propext`, `Classical.choice`, `Quot.sound` (checked by `scripts/CheckAxioms.lean`).

6 Notes on the formalisation

- **Arbitrary instead of optimal** Q: Lemmas 29, 31 and Proposition 32 hold for every $Q \in \mathrm{DS}(V, W)$ whose supported local costs are $\le L$; Theorem 9 follows by continuity, since d_1^{OT} is an infimum in Lean.
- The paper splits into $L = 0$ and $L > 0$ (then $L \ge 1$ by Lemma 29); the Lean splits into $L < 1$ and $L \ge 1$.
- The number of closing edges is bounded by counting the $\le n/\ell$ cut positions plus the m path ends instead of $\sum_j \lceil p_j/\ell \rceil$; same bound. The bound $2(k-1)L$ is used as $\le 2\ell L$, as in the paper's claim.
- Lemma 31 treats $n = 0$ separately.

7 How to check this result

- Build: `lake build GsimOT.Theorem9_LocalEditUpperBound.Theorem9` (or `bash build.sh`).
- Axioms: `#print axioms GsimOT.theorem9` (and `GsimOT.theorem9_power`); expected: `propext`, `Classical.choice`, `Quot.sound`. `CVZRounding` appears as a hypothesis in the type, not as an axiom.
- `Theorem9_correspondence.pdf` in this folder lists every paper item with its Lean declaration.